

Language sensitivity is not directional control

Matched language interventions reveal a geometry-dependent competence asymmetry in robot policies.

Research question

Does changing only a static spatial instruction reliably redirect a robot policy, and does that response imply competent task completion?

LEFT condition: "Put the Rubik's cube to the left of the bowl."

RIGHT condition: "Put the Rubik's cube to the right of the bowl."

Answer

Executed actions and endpoints often changed with the instruction, but binary task success remained direction-, layout-, and checkpoint-dependent; an exploratory block also showed descriptive variation across four prompt forms. Position reflection and a seven-level lateral sweep show that scene geometry changes the directional advantage. The effect therefore cannot be summarized as either language control or a fixed model bias.

12.4 cm

Nano RIGHT-minus-LEFT requested-depth gap; exact $p=0.000311$.

+1.12 m/m

Dose-response slope across reference-object position; 95% CI [0.72, 1.56].

$p = 0.00172$

DreamZero's failure composition differs by requested direction; a rate-only difficulty account is insufficient there.

$\rho = 0.10$

No monotonic success-gap association is detected across five DROID checkpoints; exact $p=0.95$.

DROID/RoboLab and RoboTwin use different tasks and success predicates. They are reported separately and never pooled.

Binary completion can conceal a continuous placement asymmetry

Four measurements on Cosmos3 Nano distinguish a near-ceiling task result from a directional quality gap.

Near-ceiling outcomes can hide a directional placement asymmetry

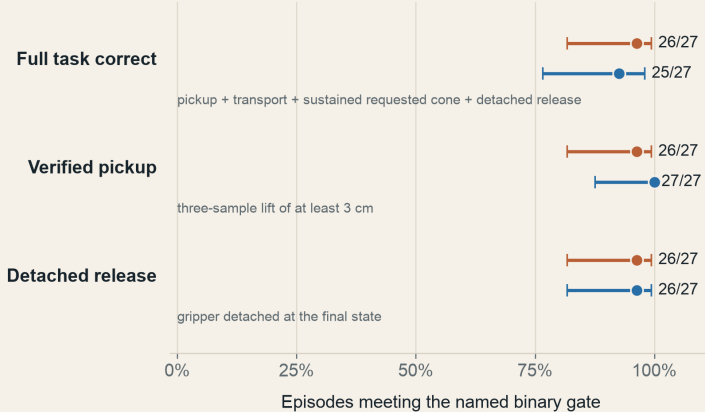
Cosmos3 Nano Policy DROID · 27 matched seeds · identical reset within each LEFT/RIGHT pair · all behavioral failures retained

LEFT prompt: "Put the Rubik's cube to the left of the bowl." RIGHT prompt: "Put the Rubik's cube to the right of the bowl."

● Prompt asks LEFT ● Prompt asks RIGHT

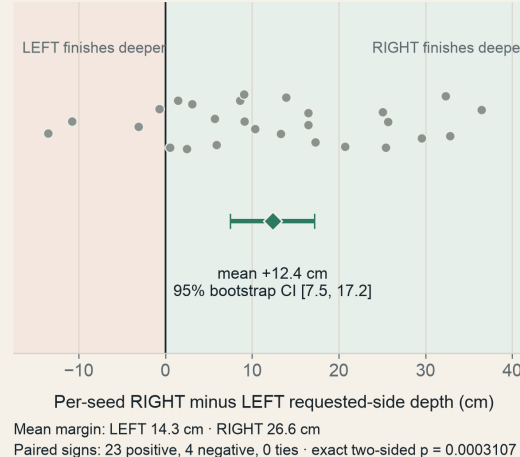
Three binary gates are near ceiling

Marginal proportions with 95% Wilson intervals



Continuous depth exposes the asymmetry

Every dot is one matched seed; failures are retained



Instrument choice changes the inference: binary gates are saturated, while signed requested-side depth preserves within-pair variation for every episode. Depth measures placement quality, not task success.

LEFT condition: "Put the Rubik's cube to the left of the bowl."

RIGHT condition: "Put the Rubik's cube to the right of the bowl."

FINDING

LEFT and RIGHT task success is nearly balanced, yet RIGHT placement is 12.4 cm deeper on average across 27 matched seeds (exact $p=0.000311$).

INTERPRETATION

Requested-side depth is the sensitive instrument at binary ceiling. It measures how deeply the object enters the requested region and remains defined on failures.

CLAIM BOUNDARY

The figure establishes a checkpoint-specific directional quality gap, not its cause. Position reflection and the dose-response sweep test geometry directly.

Position reflection shifts the directional depth advantage

The same 27 seeds and four cells are repeated on Nano, $\pi 0.5$, and DreamZero.

Position reflection shifts the directional depth advantage in all three checkpoints

Three DROID checkpoints · same 27 prespecified seeds · 108 valid episodes per checkpoint · robot base, cameras, prompts, and non-movable geometry held fixed

LEFT: "Put the Rubik's cube to the left of the bowl." RIGHT: "Put the Rubik's cube to the right of the bowl."

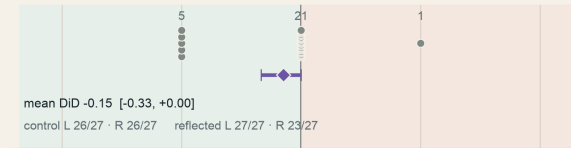
Negative interaction = reflection shifts the RIGHT-minus-LEFT advantage toward LEFT. Dots are matched seeds; diamonds are means with 95% bootstrap intervals.

REQUESTED-SIDE DEPTH INTERACTION

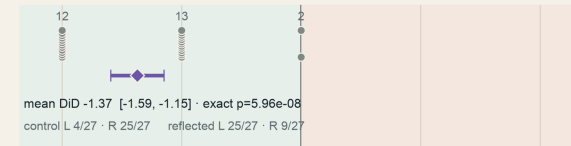
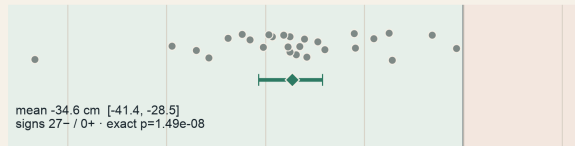
Cosmos3 Nano Policy DROID



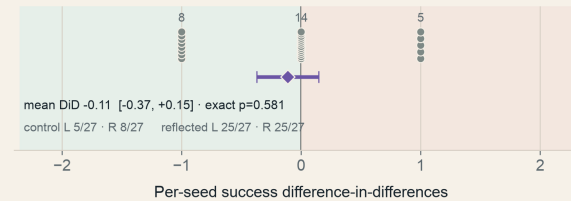
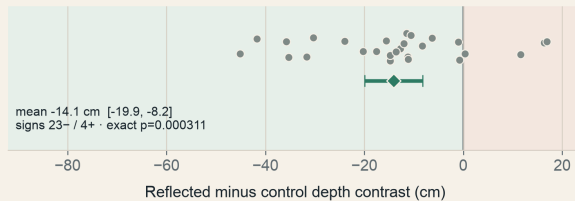
BINARY SUCCESS INTERACTION



$\pi 0.5$ DROID



DreamZero DROID



Interpretation: the position-only intervention changes which requested direction is geometrically advantaged. It does not isolate training distribution, embodiment handedness, or a full-scene symmetry transform.

LEFT condition: "Put the Rubik's cube to the left of the bowl."

RIGHT condition: "Put the Rubik's cube to the right of the bowl."

FINDING

Nano and $\pi 0.5$ show strongly negative depth interactions after reflecting movable-object positions. DreamZero depth interaction -14.1 cm; exact sign-test p=0.000311.

INTERPRETATION

Control L/R 5/27 and 8/27; reflected L/R 25/27 and 25/27. Across three checkpoints, the directional competence gap is layout-dependent rather than a fixed property of the words alone.

CLAIM BOUNDARY

The robot base, cameras, prompts, and non-movable geometry are fixed. The experiment does not separate embodiment handedness or identify training distribution.

FIGURE 3

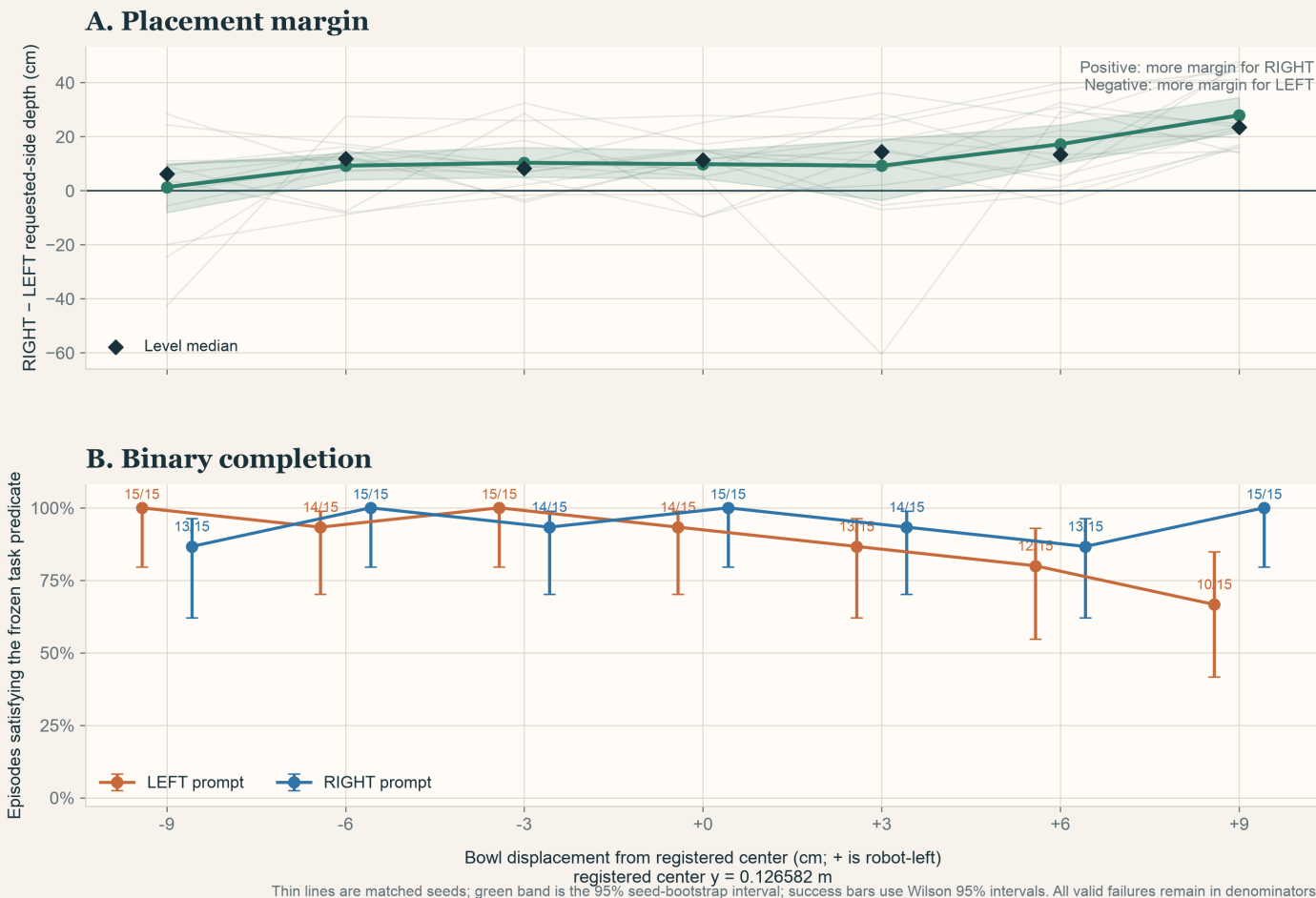
Seven reference positions reveal a positive dose-response

A seven-level sweep converts the reflection result from a binary reversal into a dose-response relationship.

Seven reference-object positions reveal a positive depth response

Cosmos3 Nano Policy DROID · 15 matched seeds at each of seven bowl positions · 210 valid episodes

LEFT: "Put the Rubik's cube to the left of the bowl." RIGHT: "Put the Rubik's cube to the right of the bowl."



LEFT condition: "Put the Rubik's cube to the left of the bowl."

RIGHT condition: "Put the Rubik's cube to the right of the bowl."

FINDING

The requested-depth gap changes by 1.12 m per meter of bowl displacement (95% bootstrap CI [0.72, 1.56]); 13/15 seed-level slopes are positive, exact $p=0.00739$.

INTERPRETATION

Across the seven sampled positions, relocating the reference object and requested regions changes the depth contrast. Binary completion is less sensitive because Nano is close to ceiling.

CLAIM BOUNDARY

The fitted zero crossing lies outside the registered support, so no in-support crossing is claimed. The curve is descriptive over the seven tested positions.

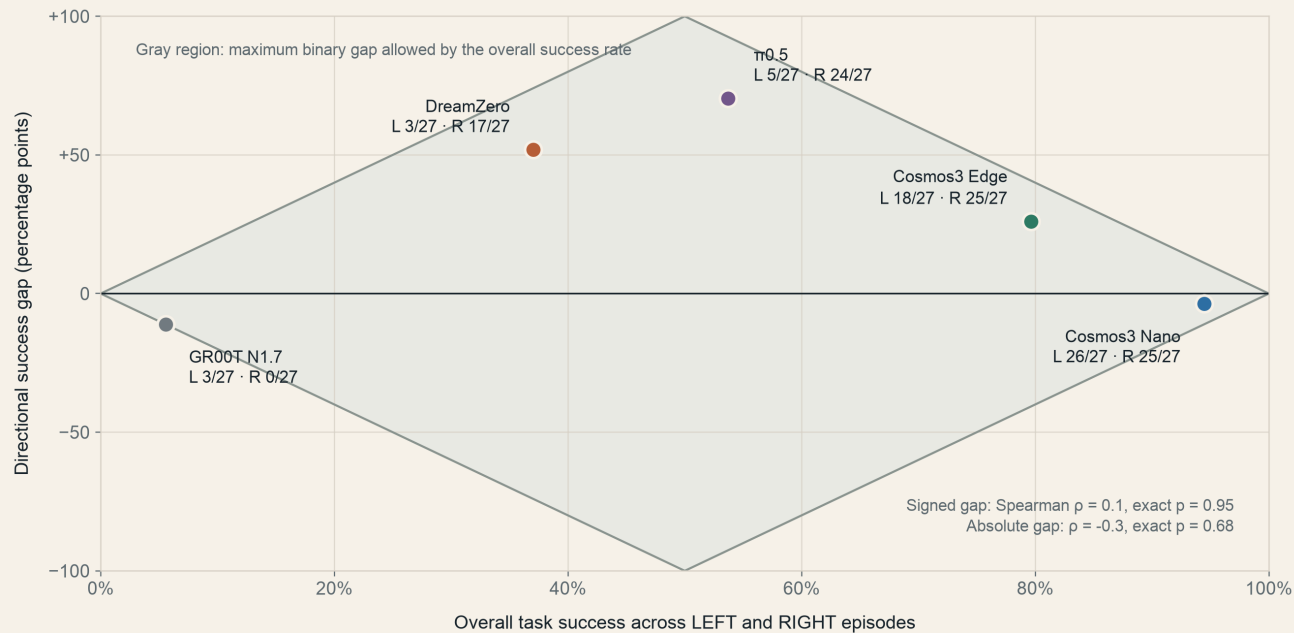
FIGURE 4

No monotonic gap–competence relation is detected

Five separate 54-episode DROID cohorts test the proposed difficulty × competence explanation descriptively.

Observed directional gaps are non-monotonic across five checkpoints

Five separate 54-episode DROID cohorts · 27 matched seeds per checkpoint · directional gap = RIGHT success rate – LEFT success rate



No monotonic association is detected in this five-checkpoint descriptive sample. Floor and ceiling compress the observable binary gap.

FINDING

There is no monotonic association with the signed gap (Spearman $\rho = 0.10$, exact $p = 0.95$) or its magnitude ($\rho = -0.30$, $p = 0.68$).

INTERPRETATION

Floor and ceiling mechanically limit the largest observable binary gap. Across these five fixed checkpoints, the observed gaps are non-monotonic; this does not identify their mechanism.

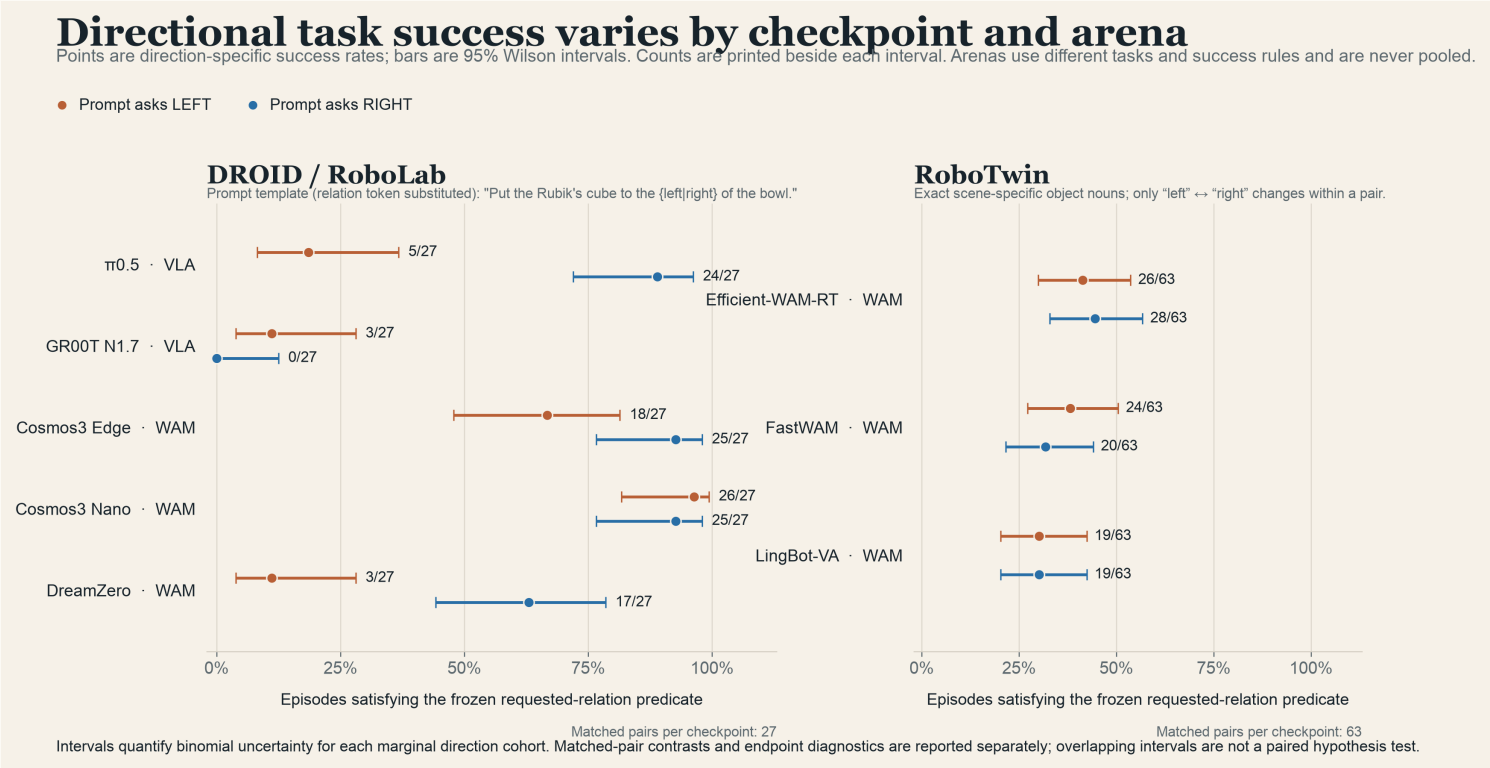
CLAIM BOUNDARY

Five checkpoints are too few for a population-level model. The figure is a descriptive constraint, not evidence that competence is irrelevant.

FIGURE 5

Directional success remains checkpoint- and arena-specific

Success counts and Wilson intervals provide context for the matched continuous diagnostics.



DROID direct prompt: "Put the Rubik's cube to the {left|right} of the bowl."
RoboTwin template: "Put the <movable object> to the {left|right} of the <reference object>." The registry fixes all seven scene-specific noun pairs.

FINDING

DROID checkpoints span near-floor to near-ceiling performance and include both positive and negative directional gaps. RoboTwin likewise varies across the three WAMs.

INTERPRETATION

These data do not support a family-level ranking. The meaningful unit is a checkpoint, arena, prompt, and frozen success predicate.

CLAIM BOUNDARY

DROID/RoboLab and RoboTwin have different tasks and success rules. Their rates are faceted and never pooled; Wilson intervals are marginal, not paired tests.

FIGURE 6

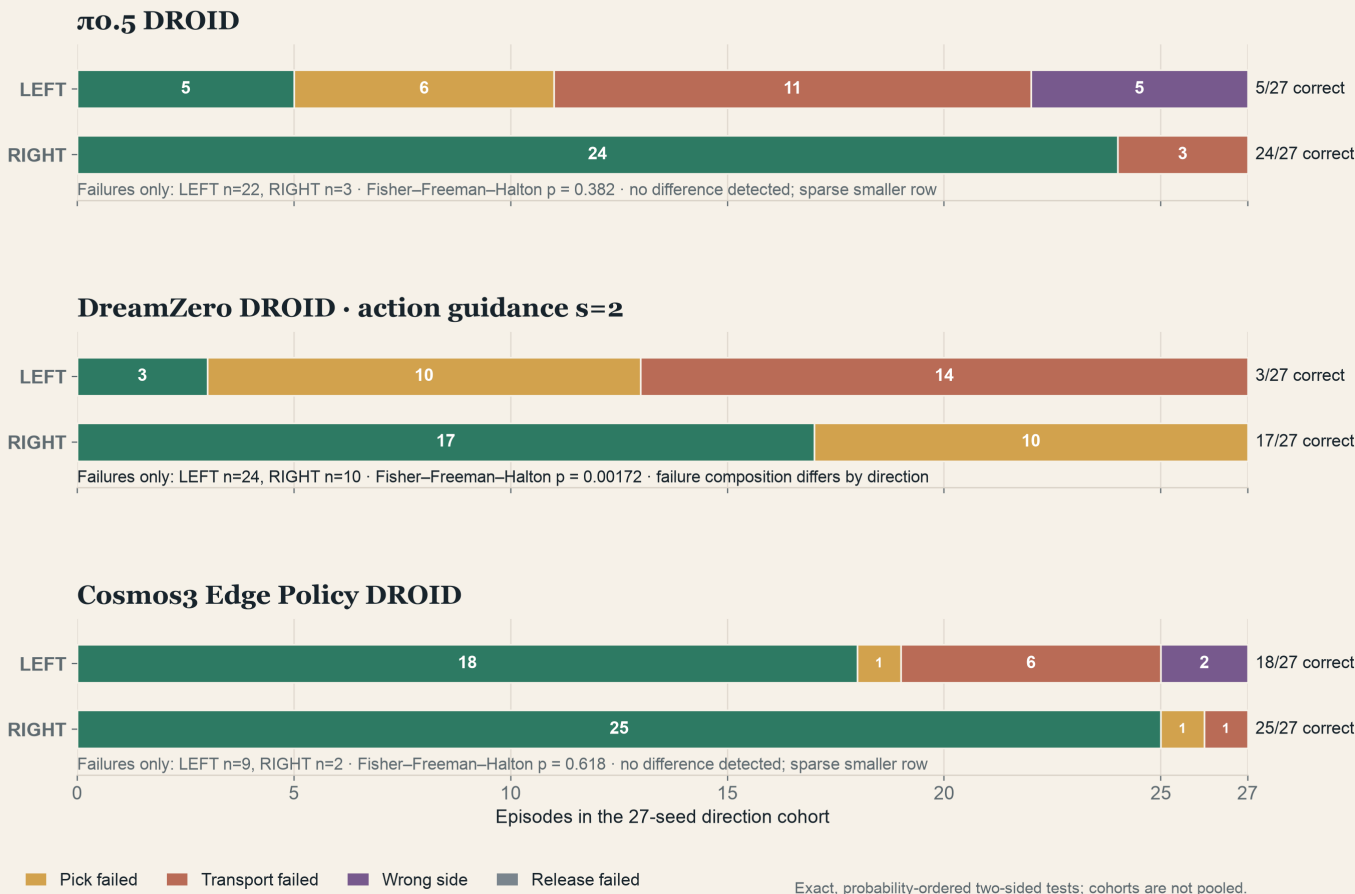
Difficulty alone does not explain every checkpoint's failures

Failure-only contingency tests ask whether LEFT and RIGHT differ only in rate or also in failure composition.

Direction changes more than the failure rate for DreamZero

Each bar is one complete 27-episode direction cohort. The exact test compares the composition of failures only; correct episodes are retained here to show competence.

LEFT prompt: "Put the Rubik's cube to the left of the bowl." RIGHT prompt: "Put the Rubik's cube to the right of the bowl."



LEFT condition: "Put the Rubik's cube to the left of the bowl."
RIGHT condition: "Put the Rubik's cube to the right of the bowl."

FINDING

$\pi 0.5$ and Edge show no detected shape difference ($p=0.382$ and $p=0.618$), but DreamZero does ($p=0.00172$).

INTERPRETATION

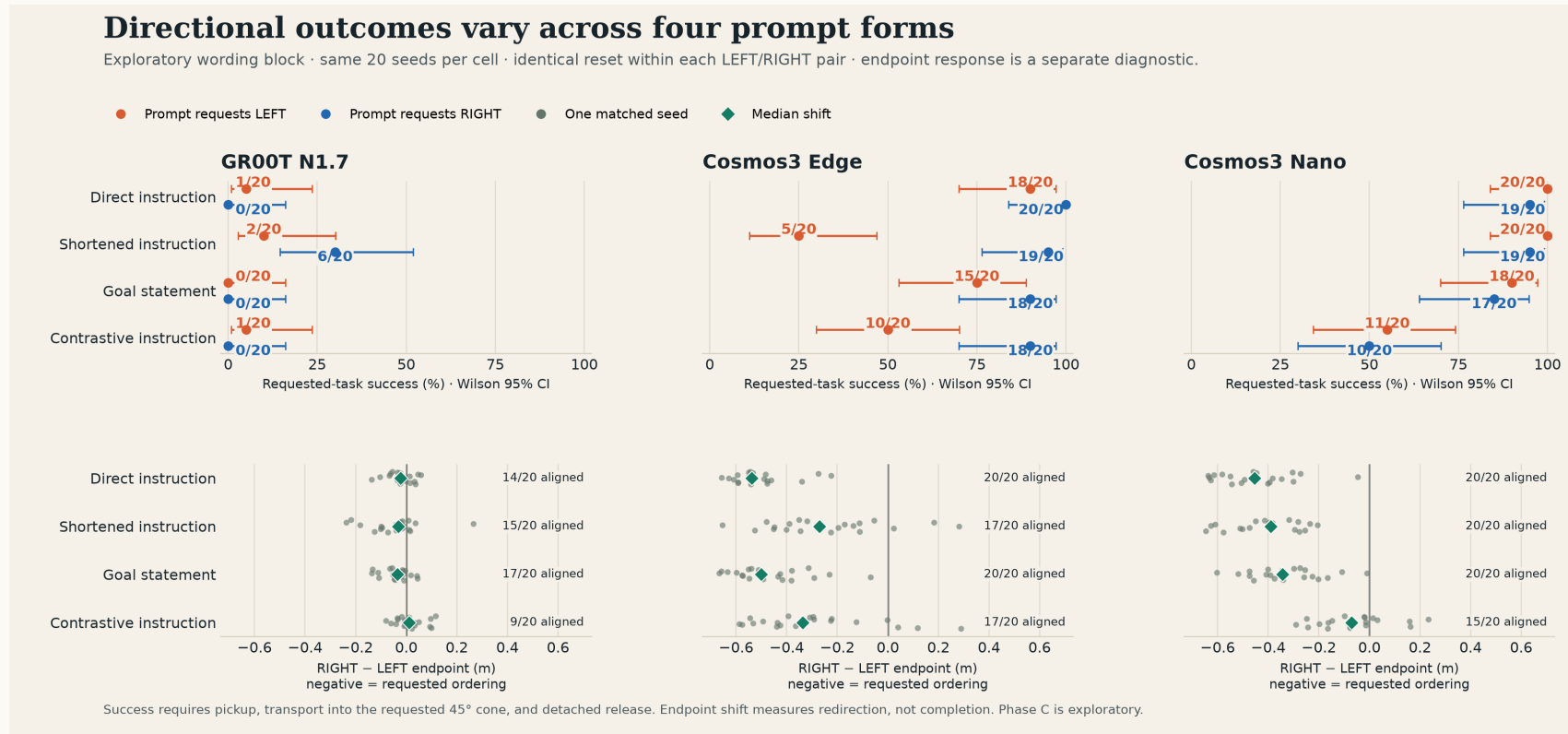
A same-shape/different-rate difficulty account remains compatible with $\pi 0.5$ and Edge but is insufficient for DreamZero, whose LEFT failures are transport-heavy while RIGHT failures are pick failures.

CLAIM BOUNDARY

Nondetection is not equivalence; the smaller direction-specific failure rows are sparse. The exact tests compare marginal failure composition, not a paired transition model.

Directional outcomes vary across four prompt forms

Exploratory: four frozen static prompt forms are crossed with direction at 20 matched seeds on three checkpoints.



THE FOUR STATIC PROMPT TEMPLATES

Direct: "Put the Rubik's cube to the {left|right} of the bowl."

Shortened: "Put the cube {left|right} of the bowl."

Goal: "The Rubik's cube should end up to the {left|right} of the bowl."

Contrastive: "Put the Rubik's cube to the {left|right} of the bowl, not to the {right|left} of the bowl."

READING

Finding: The largest descriptive direction gap is Edge under the shortened prompt: LEFT 5/20 versus RIGHT 19/20 (+14 episodes).

Interpretation: Outcomes differ descriptively across prompt forms; this is compatible with a language contribution layered over geometry, not a confirmatory phrasing effect.

Boundary: The same 20 seeds are reused; redirection and completion remain separate.

Complete checkpoint table · DROID / RoboLab

Expanded direct-command cohorts and the separate π 0-FAST compatibility cohort; no RoboTwin row is pooled here.

Checkpoint / cohort	Type	Episodes	LEFT	RIGHT	Aligned	Actions	Mean / median shift	C/P/T/W/R	Future interface
π 0.5 current stack	VLA	54	5/27	24/27	25/27	27/27	+19.2 / +20.9 cm	29/6/14/5/0	action-only
GR00T N1.7	VLA	54	3/27	0/27	21/27	27/27	+4.1 / +3.6 cm	3/49/2/0/0	action-only
Cosmos3 Edge	WAM	54	18/27	25/27	27/27	27/27	+41.4 / +44.1 cm	43/2/7/2/0	452 decoded local futures
Cosmos3 Nano	WAM	54	26/27	25/27	27/27	27/27	+40.9 / +41.2 cm	51/0/1/0/2	349 decoded local futures
DreamZero	WAM	54	3/27	17/27	25/27	27/27	+14.6 / +15.8 cm	20/20/14/0/0	54 full-reset decodes + latent futures
π 0-FAST public compatibility	VLA	40	0/20	12/20	16/20	20/20	+16.6 / +19.7 cm	12/22/4/0/2	action-only

Reading: shift is requested-oriented endpoint redirection; positive follows the requested LEFT→RIGHT ordering. C/P/T/W/R = correct, pick, transport, wrong-side, release. The older frozen π 0-FAST 20-episode reference remains a separate historical cohort and is not merged with public-compatibility evidence.

Complete checkpoint table · RoboTwin

Expanded 63-pair replicate cohorts and two bounded references; DROID success rules are not used here.

Checkpoint / cohort	Type	Episodes	LEFT	RIGHT	Aligned	Actions	Mean / median shift	C/P/T/W/R	Future interface
Efficient-WAM-RT	WAM	126	26/63	28/63	42/63	63/63	-14.4 / +4.4 cm	54/28/38/6/0	126 decoded coarse futures
FastWAM	WAM	126	24/63	20/63	39/63	63/63	-0.4 / +1.8 cm	44/31/36/15/0	action-only at test time
LingBot-VA	WAM	126	19/63	19/63	47/63	63/63	+19.9 / +11.0 cm	38/9/64/15/0	126 latent-only futures
BOUNDED V2: LingBot-VLA 4B	VLA	6	1/3	0/3	2/3	NR	NR	not harmonized	none
BOUNDED V2: Light-WAM	WAM	6	1/3	0/3	1/3	3/3	NR	not harmonized	action only infer action

Reading: shift is requested-oriented endpoint redirection; positive follows the requested LEFT→RIGHT ordering. C/P/T/W/R = correct, pick, transport, wrong-side, release. Efficient-WAM-RT's negative mean but positive median reflects a small reverse-tail of large shifts, not a transcription error. Nesting: each expanded WAM has 63 pairs from nine policy seeds within seven scenes. NR is never converted to zero.

Registered controls refine—and limit—the mechanism claim

The controls below test stochasticity, scene factors, and cross-arena generality without pooling their estimands.

ROBOTWIN POSITION REFLECTION

FastWAM reverses under reflected movable-object positions

Requested-depth interaction -0.76 m, 95% CI [-0.84, -0.69], exact sign $p=1.49\text{e-}08$. Binary success DiD -0.48, exact $p=0.000488$.

Boundary: This is cross-arena replication of geometry dependence, not a pooled success estimate: RoboTwin uses a different controller, scenes, and success predicate.

TARGET START-SIDE

The success advantage follows the target's initial side

Target starts LEFT: L 26/27, R 22/27. Target starts RIGHT: L 23/27, R 27/27. The outer binary-gap interaction is +0.296 (exact $p=0.0156$). Depth changes by +0.079 m, 95% CI [0.012, 0.147].

Boundary: The binary interaction is clear; the continuous-depth mean CI excludes zero but its sign test is $p=0.122$, so sign consistency is not claimed. Start side changes geometry, reachability, and policy state together.

TARGET / REFERENCE ROLE SWAP

Object roles alter the magnitude of directional steering

Swapping cube and bowl roles changes requested-depth contrast by +0.13 m (95% CI [0.06, 0.21]; $p=0.00151$). The binary interaction is +0.30, exact $p=0.0557$.

Boundary: Continuous redirection changes clearly; the binary interaction narrowly misses 0.05. Object semantics and physical affordances change together, so this is not language-only evidence.

FIXED-SCENE STOCHASTIC REPEATS

Repeated policy samples separate scene effects from policy noise

Across 27 fixed scenes $\times$ 8 policy samples per direction, LEFT succeeds 41/216 and RIGHT 197/216. The seed-level mean $p(\text{RIGHT})-p(\text{LEFT})$ is +0.72 (95% cluster-bootstrap CI [0.68, 0.77]).

Boundary: The environment seed is the inferential unit; 432 episodes are nested repeats, not 432 independent scenes. Wilson intervals over episodes are descriptive only.

Base-rotation control: failed closed before behavioral inference because the wrist camera inherits the robot-base transform, violating the registered fixed-camera contract. No behavioral denominator or null result was created.

Methods, inference, and claim boundary

Frozen definitions make the expanded and intervention cohorts comparable without erasing their different scopes.

DESIGN

Matched seed and identical reset within each language pair. Static episode prompt only. No oracle, subtask coach, prompt switching, or progress-conditioned language. Position reflection moves movable-object centers only; the lateral sweep moves the reference object across seven registered positions.

OUTCOMES

For DROID/RoboLab, correct requires pickup, transport into the requested 45° cone, sustained entry, and detached release. RoboTwin retains its own frozen arena predicate. Continuous fields include signed lateral offset, requested-side depth, timing, and path diagnostics where exposed.

INFERENCE

Marginal proportions use Wilson intervals. Continuous matched contrasts use 10,000–20,000 seed-level bootstrap resamples, as preregistered per cohort, plus exact two-sided sign tests. Binary reflection effects use exact within-seed layout-label permutation. Phase C is exploratory.

What the evidence supports

Language frequently perturbs executed behavior. Geometry can reverse the directional competence advantage. Outcomes also vary descriptively across prompt forms in the exploratory block. Sensitivity, requested endpoint ordering, and task completion are therefore distinct estimands.

What it does not identify

The interventions do not identify training-distribution frequency, a unique causal source inside a checkpoint, or a universal VLA–WAM family effect. Robot-base rotation was structurally invalid because the wrist camera inherits the base transform.

Evidence handling

Behavioral failures remain in denominators. Infrastructure-invalid and partial attempts are separate. Raw rollouts, videos, and checkpoints remain on the ali-owned PVC; compact JSONL, hashes, manifests, code, and selected media are committed.